

RAJNEESH ANAND

Email: raa524@lehigh.edu | Phone: (610)-570-3908 | Github: [rajneeshanand](#)

RESEARCH INTERESTS

Optimal Control, Microchemical Systems, Computational Fluid Dynamics, Transport Processes, and Soft Matter

EDUCATION & QUALIFICATION

PhD, Chemical Engineering

Lehigh University, Bethlehem, PA, USA

Aug. 2024 – Present

(Current GPA: 3.96/4.0)

- Advisor: Dr. Mayuresh V. Kothare
- Currently working on the intersection of machine learning, optimal control, and microfluidics

Bachelor of Technology (B.Tech), Chemical Engineering

National Institute of Technology Durgapur, WB, India

2019 – 2023

(CGPA: 8.30/10)

- Advisors: Dr. Abhiram Hens and Dr. Mrinal Kanti Mandal
- Thesis: *Optimizing the design and surface wettability of desalination-based solar stills*
- First Class with Distinction
- Graduate Aptitude Test in Engineering (GATE – Chemical Engg.): Qualified [2022 & 2023 (94th percentile)]

SCIENTIFIC JOURNAL PUBLICATIONS

- [1] **Anand, R.**, Kothare, M.V. Optimal control and stability of droplet transport in a microchannel. (*Manuscript under preparation*)
- [2] **Anand, R.**, Kothare, M.V. Choreography of a droplet: passive guiding, holding, and surfactant-free splitting. (*Manuscript under preparation*)
- [3] **Anand, R.**, Mandal, M.K., Chaudhuri, R.G., Hens, A. (2024). Effect of patterned condensation surface on the performance of a newly designed basin-type solar desalination unit. *Applied Thermal Engineering*.
- [4] **Anand, R.**, Mandal, M.K., Chaudhuri, R.G., Hens, A. (2023). Optimization of geometry and surface wettability for enhanced efficiency of the solar still-based desalination unit. *Int. Journal of Ambient Energy*.

AWARDS

Third Prize (National Level) in the **New Generation Ideation Contest 2022 (NGIC)**, issued by HP Green R&D Centre (HPCL, India), awarded on 17th March 2023.

Topic: Bi-layered ceramic membrane single separator-photo-reactor for the capture & conversion of CO₂ to CH₄.

CONFERENCE PROCEEDINGS & PRESENTATIONS

- [1] **Anand, R.**, Kothare, M.V. (2026). Optimal control of droplet transport in a microchannel. *IEEE Conference on Decision and Control (CDC) 2026* (Under submission).
- [2] **Anand, R.**, Kothare, M.V. (2026). Surfactant-free droplet breakup dynamics under confinement. *APS Global Physics Summit (March Meeting) 2026*.
- [3] **Anand, R.**, Kumar, I., Hens, A. (2023). Effect of bed porosity and inlet temperature on hydrogen adsorption inside a cylindrical storage tank: A CFD study. *IICHE-CHEMCON 2023*.
- [4] Bhattacharya, M., Akhtar, F., **Anand, R.** (2023). Numerical investigation of heat transfer enhancement in heat exchangers using CFD. *IICHE-CHEMCON 2023*.
- [5] **Anand, R.**, Ray, R., Akhtar, F. (2023). Modeling and simulation of cumene process in Aspen Plus. *Int. Conf. on Recent Advances in Technology, Engineering, Science and Management (ICRATESM-2023), Book Chapter*.

- [6] Akhtar, F., **Anand, R.**, Hens, A. (2023). Computational study of pipeline transportation of crude oil. *Int. Conference on Chemical Engineering Innovations and Sustainability (ICEIS-2023)*, Jadavpur University.
- [7] Akhtar, F., **Anand, R.** (2023). Computational study of flow past a cylinder in tandem setup. *4th Int. Conference on Recent Advancements in Mechanical Engineering (ICRAME-2023)*. (Springer Nature).
- [8] Akhtar, F., & **Anand, R.** (2023). The adsorptive hydrogen storage in an activated carbon tank based on a modified Dubinin–Astakhov model. *Int. Conference on Materials Science and Mechanical Engineering (ICMSME-2023)* (Book Chapter).
- [9] **Anand, R.**, Akhtar, F. (2022). Effect of operating parameters on nickel removal in a fixed bed adsorption column. *Int. Conference on Advances and Creations in Mechanical Engineering (ICACME)*.

WORK EXPERIENCE

Graduate Engineer Trainee (Process Engineer)

Aug 2023 – Dec 2023

Reliance Industries Ltd., Jamnagar, India

- Worked on mathematical modeling and simulation of a diolefin saturation reactor used in the DTA Coker Naphtha Hydrotreater unit (CNHT) of the Jamnagar (RIL) Complex.
- Worked as a Technologist (in CTS) during shutdown in the SEZ Coker unit of Jamnagar (RIL) Complex.

Industrial Engineering Intern

May – June, 2022

Mitsubishi Chemical Corporation, Haldia, WB, India

- Analyzed azeotropic and binary distillation: tray performance evaluation and checklist preparation.
- Studied project layout, PFDs and P&IDs, safety procedures, and the function of each unit operation, including thermodynamics and phase equilibrium in distillation columns.

B.TECH PROJECTS & RESEARCH

Flow Past a Cylinder in Tandem Arrangement

2022 – 2023

- Studied the impact of Reynolds number and gap spacing on flow patterns; showed that drag coefficients of both cylinders in a tandem setup change rapidly at a critical separation distance.

Hydrogen Adsorption in a Packed Bed of Activated Carbon

2022 – 2023

- Investigated the influence of bed porosity and inlet temperature on hydrogen adsorption performance.

Final Year Project: Solar Desalination Unit

2022 – 2023

- Studied the influence of design and surface wettability on the condensing plate.
- Proposed a novel desalination unit with patterned surface design to enhance solar still efficiency.

Shell and Tube Heat Exchanger: A CFD Study

2021 – 2022

- Conducted CFD analysis of shell and tube heat exchangers considering different flow arrangements, geometric influence, and baffle shapes.

TECHNICAL SKILLS

CFD Software:	ANSYS Fluent, COMSOL Multiphysics, FEniCS
Programming:	MATLAB, Python, Parallel Computing & HPC
Numerical Methods:	FEM, FVM, PDE Solvers, Continuous (Nonlinear) Optimization
Process Simulation:	Aspen Plus